1 Gain law

(bars: lengths by R; $a_o := \Delta t/(\gamma_N R)$, $a_{\text{nom}} := a_o R$, $\bar a := a/a_{\text{nom}}$, $\bar f := a_o f$, $\bar a\,\bar f = \Delta\bar w$; $e_\theta = \theta - \hat\theta$; $\mathbb{E}[\cdot]$ over the noise; $P_{ij}[k] = \mathbb{E}[e_i[k]\,e_j[k]]$ posterior at k, P^f forecast; $\bar w_d[k]$ known)

$$\bar a_{\text{true}}(\bar w) = 1 - \frac{9}{8\bar w} + \frac{1}{2\bar w^3} - \frac{57}{100\,\bar w^4} + \frac{1}{5\bar w^5} + \frac{7}{200\,\bar w^{11}} - \frac{1}{25\bar w^{12}}$$

$$\begin{aligned}
\bar a'_{\text{true}}(\bar w) &= \frac{9}{8\bar w^2} - \frac{3}{2\bar w^4} + \frac{57}{25\,\bar w^5} - \frac{1}{\bar w^6} - \frac{77}{200\,\bar w^{12}} + \frac{12}{25\,\bar w^{13}}\\
\bar a''_{\text{true}}(\bar w) &= -\frac{9}{4\bar w^3} + \frac{6}{\bar w^5} - \frac{57}{5\,\bar w^6} + \frac{6}{\bar w^7} + \frac{231}{50\,\bar w^{13}} - \frac{156}{25\,\bar w^{14}}
\end{aligned}$$

$$\begin{gathered}
\bar a'[k] := \bar a'_{\text{true}}\big(\bar w_d[k] - \delta\hat{\bar w}_3[k]\big), \qquad \bar a''[k] := \bar a''_{\text{true}}\big(\bar w_d[k] - \delta\hat{\bar w}_3[k]\big)\\
\frac{\partial \bar a'[k]}{\partial \bar a_w} = 0, \qquad \frac{\partial \bar a'[k]}{\partial \delta\bar w_3} = -\bar a''[k], \qquad a_w = a_o\,\bar a_w
\end{gathered}$$

2 Jacobians

(row 4 of F_e, first order in the step; the exact-map corrections, relative size $\bar a'[\Delta\bar w_d + (1-\lambda_c)\delta\hat{\bar w}_3]/(1-\hat{\bar a}_w) \approx 10^{-2}$, are not carried)

$$\begin{aligned}
F_e(4,3) &= (1-\lambda_c)\,\bar a'[k] - \bar a''[k]\big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\big]\\
F_e(4,4) &= 1 + F_{dw}[k]\,\bar a'[k]
\end{aligned}$$

3 True dynamics

$\big(\delta\bar w_j[k] := \delta\bar w[k-d+j-1],\ j = 1,\dots,d+1,\ d = 2\big)$

$$\begin{aligned}
\bar w[k] &= \bar w_d[k] - \delta\bar w_3[k]\\
\bar w_d[k+1] &= \bar w_d[k] + \Delta\bar w_d[k]\\
\delta\bar w_1[k+1] &= \delta\bar w_2[k]\\
\delta\bar w_2[k+1] &= \delta\bar w_3[k]\\
\delta\bar w_3[k+1] &= \lambda_c\,\delta\bar w_3[k] - F_{dw}[k]\,e_{\bar a_w}[k] - \varepsilon_w[k]\\
\bar w[k+1] - \bar w[k] &= \Delta\bar w_d[k] + (1-\lambda_c)\,\delta\bar w_3[k] + F_{dw}[k]\,e_{\bar a_w}[k] + \varepsilon_w[k]\\
\bar a_w[k+1] &= \bar a_{\text{true}}\big(\bar w[k+1]\big)
\end{aligned}$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k-i]$$

$$\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]$$

4 Process model

$$\mathbf{x} = \begin{bmatrix} \delta\bar{w}_1, & \delta\bar{w}_2, & \delta\bar{w}_3, & \bar{a}_w \end{bmatrix}^\top \qquad (d=2)$$

5 Estimator

$$\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \bar{a}'[k]\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big] \\
&\quad + \tfrac{1}{2}\,\bar{a}''[k]\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big]^2 \\
&\quad + \tfrac{1}{2}\,\bar{a}''[k]\Big[(1-\lambda_c)^2 P_{33}[k] + 2(1-\lambda_c)\,F_{dw}[k]\,P_{34}[k] + F_{dw}[k]^2 P_{44}[k] + Q_{33}[k]\Big] \\
&\quad - \bar{a}''[k]\Big[(1-\lambda_c)\,P_{33}[k] + F_{dw}[k]\,P_{34}[k]\Big]
\end{aligned}$$

($\ell_{31}[k]$: entry $(3,1)$ of $L[k]$, the y_1 gain of the update at k; Measurement section. The n_w feedthrough adds no line: its four shares cancel, Error dynamics section.)

code: the first two terms of the $\hat{\bar{a}}_w$ row are carried by the exact step,

$$\begin{aligned}
&\frac{\bar{a}'[k]\big(1-\hat{\bar{a}}_w[k]\big)\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big]}{\big(1-\hat{\bar{a}}_w[k]\big) + \bar{a}'[k]\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big]} \\
&\qquad = \bar{a}'[k]\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big] - \frac{\bar{a}'[k]^2}{1-\hat{\bar{a}}_w[k]}\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big]^2 + \mathcal{O}(3)
\end{aligned}$$

whose second-order coefficient is $-\bar{a}'^2/(1-\hat{\bar{a}}_w)$ instead of $\frac{1}{2}\bar{a}''$; the difference $\frac{1}{2}\big(\bar{a}''[k] + 2\bar{a}'[k]^2/(1-\hat{\bar{a}}_w[k])\big)[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]]^2$ is added together with the two mean lines (pred_mean2).

6 Error dynamics

(True − Estimator)

$$
\begin{aligned}
e_{\delta\bar w_1}[k+1] &= e_{\delta\bar w_2}[k]\\
e_{\delta\bar w_2}[k+1] &= e_{\delta\bar w_3}[k]\\
e_{\delta\bar w_3}[k+1] &= \lambda_c\, e_{\delta\bar w_3}[k] - F_{dw}[k]\, e_{\bar a_w}[k] - \varepsilon_w[k]\\
e_{\bar a_w}[k+1] &= \big[1 + F_{dw}[k]\,\bar a'[k]\big]\, e_{\bar a_w}[k] + \Big[(1-\lambda_c)\,\bar a'[k] - \bar a''[k]\big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\big]\Big]\, e_{\delta\bar w_3}[k]\\
&\qquad + \bar a'[k]\,\varepsilon_w[k] + (\text{second order, below})
\end{aligned}
$$

$$
F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & \lambda_c & -F_{dw}\\ 0 & 0 & (1-\lambda_c)\,\bar a' - \bar a''\big[\Delta\bar w_d + (1-\lambda_c)\,\delta\hat{\bar w}_3\big] & 1 + F_{dw}\,\bar a' \end{bmatrix}, \qquad \mathbf{q}[k] = \Big[0,\ 0,\ -\varepsilon_w,\ \bar a'[k]\,\varepsilon_w\Big]^\top
$$

known and unknown parts of the true step, and the two starting points

$$
\begin{aligned}
\bar w[k+1] - \bar w[k] &= \Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\Big] + \Big[(1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]\\
\bar w[k] &= \big(\bar w_d[k] - \delta\hat{\bar w}_3[k]\big) - e_{\delta\bar w_3}[k]
\end{aligned}
$$

truth, second order about $\bar w_d[k] - \delta\hat{\bar w}_3[k]$

$$
\begin{aligned}
\bar a'_{\text{true}}\big(\bar w[k]\big) &= \bar a'[k] - \bar a''[k]\, e_{\delta\bar w_3}[k] + \mathcal{O}(e^2), \qquad \bar a''_{\text{true}}\big(\bar w[k]\big) = \bar a''[k] + \mathcal{O}(e)\\
\bar a_w[k+1] - \bar a_w[k] &= \bar a'_{\text{true}}\big(\bar w[k]\big)\big[\bar w[k+1] - \bar w[k]\big] + \tfrac{1}{2}\,\bar a''_{\text{true}}\big(\bar w[k]\big)\big[\bar w[k+1] - \bar w[k]\big]^2 + \mathcal{O}(3)\\
&= \bar a'[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\Big] + \bar a'[k]\Big[(1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]\\
&\quad - \bar a''[k]\, e_{\delta\bar w_3}[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\\
&\qquad\qquad + (1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]\\
&\quad + \tfrac{1}{2}\,\bar a''[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\Big]^2\\
&\quad + \bar a''[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\Big]\Big[(1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]\\
&\quad + \tfrac{1}{2}\,\bar a''[k]\Big[(1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]^2
\end{aligned}
$$

estimator, second order (first two terms of the $\hat{\bar a}_w$ row)

$$
\hat{\bar a}_w[k+1] - \hat{\bar a}_w[k] = \bar a'[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\Big] + \tfrac{1}{2}\,\bar a''[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\Big]^2
$$

truth − estimator

$$
\begin{aligned}
e_{\bar a_w}[k+1] - e_{\bar a_w}[k] &= \bar a'[k]\Big[(1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]\\
&\quad - \bar a''[k]\, e_{\delta\bar w_3}[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\\
&\qquad\qquad + (1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]\\
&\quad + \bar a''[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\hat{\bar w}_3[k]\Big]\Big[(1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]\\
&\quad + \tfrac{1}{2}\,\bar a''[k]\Big[(1-\lambda_c)\, e_{\delta\bar w_3}[k] + F_{dw}[k]\, e_{\bar a_w}[k] + \varepsilon_w[k]\Big]^2 + \mathcal{O}(3)
\end{aligned}
$$

mean, with A1 $\mathbb{E}[e_{\delta\bar{w}_3}] = \mathbb{E}[e_{\bar{a}_w}] = 0$, A2 $\mathbb{E}[\delta\hat{\bar{w}}_3\, e_{\delta\bar{w}_3}] = \mathbb{E}[\delta\hat{\bar{w}}_3\, e_{\bar{a}_w}] = 0$, A3 $\mathbb{E}[\varepsilon_w\, e_{\bar{a}_w}] = 0$, $\mathbb{E}[\varepsilon_w^2] = Q_{33}$, and (the update at k and the controller read the same $y_1[k]$: $\delta\hat{\bar{w}}_3[k] \ni \ell_{31}[k]\, n_w[k]$, $e_{\delta\bar{w}_3}[k] \ni -\ell_{31}[k]\, n_w[k]$, $\varepsilon_w[k] \ni (1-\lambda_c)\, n_w[k]$, $\bar{a}'[k] = \bar{a}'_{\text{true}}(\bar{w}_d[k] - \delta\hat{\bar{w}}_3[k]) \ni -\bar{a}''[k]\, \ell_{31}[k]\, n_w[k]$)

$$\begin{aligned}
\mathbb{E}\big[\delta\hat{\bar{w}}_3[k]\, \varepsilon_w[k]\big] &= (1-\lambda_c)\, \ell_{31}[k]\, R_1 \\
\mathbb{E}\big[e_{\delta\bar{w}_3}[k]\, \varepsilon_w[k]\big] &= -(1-\lambda_c)\, \ell_{31}[k]\, R_1 \\
\mathbb{E}\big[\bar{a}'[k]\, \varepsilon_w[k]\big] &= -\bar{a}''[k]\, (1-\lambda_c)\, \ell_{31}[k]\, R_1
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big] - \mathbb{E}\big[e_{\bar{a}_w}[k]\big] &= -\bar{a}''[k]\, (1-\lambda_c)\, \ell_{31}[k]\, R_1 \\
&\quad - \bar{a}''[k]\Big[(1-\lambda_c)\, P_{33}[k] + F_{dw}[k]\, P_{34}[k]\Big] + \bar{a}''[k]\, (1-\lambda_c)\, \ell_{31}[k]\, R_1 \\
&\quad + \bar{a}''[k]\, (1-\lambda_c)^2\, \ell_{31}[k]\, R_1 \\
&\quad + \tfrac{1}{2}\, \bar{a}''[k]\Big[(1-\lambda_c)^2 P_{33}[k] + 2(1-\lambda_c)\, F_{dw}[k]\, P_{34}[k] + F_{dw}[k]^2 P_{44}[k] + Q_{33}[k]\Big] \\
&\quad - \bar{a}''[k]\, (1-\lambda_c)^2\, \ell_{31}[k]\, R_1 \\
&= -\bar{a}''[k]\Big[(1-\lambda_c)\, P_{33}[k] + F_{dw}[k]\, P_{34}[k]\Big] \\
&\quad + \tfrac{1}{2}\, \bar{a}''[k]\Big[(1-\lambda_c)^2 P_{33}[k] + 2(1-\lambda_c)\, F_{dw}[k]\, P_{34}[k] + F_{dw}[k]^2 P_{44}[k] + Q_{33}[k]\Big]
\end{aligned}$$

(lines 1–4 in the order of the truth − estimator expansion; the four $\ell_{31} R_1$ shares sum to zero)

compensation: the two lines added to $\hat{\bar{a}}_w[k+1]$ (Estimator section)

$$\mathbb{E}\big[e_{\bar{a}_w}[k+1]\big] - \mathbb{E}\big[e_{\bar{a}_w}[k]\big] = 0$$

code: with the exact step in place of $\bar{a}'[\cdot] + \frac{1}{2}\bar{a}''[\cdot]^2$ the mean gains $\frac{1}{2}\big(\bar{a}''[k] + 2\bar{a}'[k]^2/(1-\hat{\bar{a}}_w[k])\big)[\Delta\bar{w}_d[k] + (1-\lambda_c)\, \delta\hat{\bar{w}}_3[k]]^2$, which pred_mean2 adds back.

7 Measurement and innovation

$$\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k], \qquad \mathrm{Var}(n_a) = R_2 \\
\bar{a}_w[k-d] &= \bar{a}_w[k] - \bar{a}'[k]\, \nabla_d \bar{w}_d[k] \qquad \text{(first order, as in the code)}
\end{aligned}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad L[k] = P^f H^\top \big(H P^f H^\top + R\big)^{-1}$$

$$e_{y_1}[k] = e_{\delta\bar{w}_1}[k] + n_w[k], \qquad e_{y_2}[k] = e_{\bar{a}_w}[k] + n_a[k]$$

8 Process and measurement noise

$$\kappa_T := \frac{4k_B T\, a_o}{R}, \qquad \mathrm{Var}\big(\bar a_w[k]\, \bar f_T[k]\big) = \kappa_T\, \bar a_w[k] \to \kappa_T\, \hat{\bar a}_w[k], \qquad \sigma^2_{n_w} = \frac{\sigma_n^2}{R^2}$$

$$Q_{33} = \kappa_T \Big(\hat{\bar a}_w[k] + (1-\lambda_c)^2 \big(\hat{\bar a}_w[k-1] + \hat{\bar a}_w[k-2]\big)\Big) + (1-\lambda_c)^2 \sigma^2_{n_w}$$

$$Q_{34} = Q_{43} = -\bar a'[k]\, Q_{33}, \qquad Q_{44} = \bar a'[k]^2\, Q_{33}$$

$$R = \mathrm{diag}(R_1, R_2), \qquad R_1 = \sigma^2_{n_w}, \qquad R_2 = 2a^2_{\mathrm{cov}}\, IF_{\mathrm{var}}\big(\hat{\bar a}_w\big)\, \big(\hat{\bar a}_w + \xi\big)^2$$

$$IF_{\mathrm{var}}\big(\hat{\bar a}_w\big) = 1 + 2\Big[c_1 u^2 + 2c_2\, u(1-u) + c_3(1-u)^2\Big], \qquad u = \frac{\hat{\bar a}_w}{\hat{\bar a}_w + \xi}, \qquad \xi = \rho_n\, \frac{\sigma^2_{n_w}}{\kappa_T}$$

($c_1 = 1.42044$, $c_2 = 0.0098975$, $c_3 = 0.098910$, $\rho_n = 0.350941$, $a_{\mathrm{cov}} = 0.05$)

Hold level mode

($\Delta \bar w_d = 0$, $\nabla_d \bar w_d = 0$; $\mu_3 := \mathbb{E}[e_{\delta \bar w_3}]$, $\mu_4 := \mathbb{E}[e_{\bar a_w}]$, $\iota_1 := \mathbb{E}[e_{y_1}]$, $\iota_2 := \mathbb{E}[e_{y_2}]$, $\bar F := \mathbb{E}[F_{dw}]$, $g := \mathrm{Cov}(F_{dw}, e_{\bar a_w}) + \mathbb{E}[\varepsilon_w]$, $\beta_2 :=$ readout level offset, $r :=$ predict mean residual per step, $\gamma := a_{\mathrm{cov}}(1-S)$; superscript $-$ = before the update)

$$\begin{aligned}
\mu_3^-[k+1] &= \lambda_c\, \mu_3[k] - \bar F\, \mu_4[k] - g \\
\mu_4^-[k+1] &= \mu_4[k] + \bar a'(1-\lambda_c)\, \mu_3[k] + \bar a' \bar F\, \mu_4[k] + \bar a' g + r \\
\iota_1[k+1] &= \mu_3^-[k+1] \\
\iota_2[k+1] &= a_{\mathrm{cov}}\big[(1-S)\, \mu_4^-[k+1] + \beta_2\big] \\
\mu_3[k+1] &= \mu_3^-[k+1] - \ell_{31}\, \iota_1[k+1] - \ell_{32}\, \iota_2[k+1] \\
\mu_4[k+1] &= \mu_4^-[k+1] - \ell_{41}\, \iota_1[k+1] - \ell_{42}\, \iota_2[k+1]
\end{aligned}$$

steady state ($\ell_{32} = 0$)

$$D = 1 - (1-\ell_{31})\, \lambda_c, \qquad E = \ell_{41} + \bar a'\, \ell_{31}\, (1 - \ell_{42}\gamma)$$

$$\mu_3^- = -\frac{\bar F \mu_4 + g}{D}, \qquad \mu_3 = (1-\ell_{31})\, \mu_3^-$$

$$\kappa\, \mu_4 = s, \qquad \kappa = \ell_{42}\gamma - \frac{E\, \bar F}{D}, \qquad s = \frac{E\, g}{D} + (1-\ell_{42}\gamma)\, r - \ell_{42}\, a_{\mathrm{cov}}\, \beta_2$$

$$\bar a'(1-\lambda_c)\, \mu_3 + \bar a' \bar F\, \mu_4 + \bar a' g = -\bar a'(1-\lambda_c)\, \mathbb{E}\big[\delta \hat{\bar w}_3\big] \qquad \text{(stationary hold: } \mathbb{E}[\bar a_w[k+1] - \bar a_w[k]] = 0\text{)}$$

two-step delay kept: $D \to (1+\ell_{21})(1-\lambda_c) + \ell_{31}$

$$g = r = \bar F = 0 \;\Rightarrow\; \mu_4 = -\frac{\beta_2}{1-S}$$